MLR780 Air Conditioning and Refrigeration 2021

Assignment 1
Psychrometry

Instructions

- This individual assignment is to be submitted online before midnight on **18 April 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment1-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment1-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Background:

Consider a simplistic air conditioning system contained in Figure 1. Air is drawn in from the atmosphere at point 1. It can be mixed with return air from state 6 to produced state 2, where-after it enters a cooling coil to be cooled and dehumidified via heat transfer with a chilled water loop to state 3. It then passes through the supply fan and an air duct in which it could be reheated by an electric re-heater to state 4, and then supplied to the zone. Assume the room to be occupied by office workers. The zone has both sensible ($\dot{Q}_S$) and latent cooling ($\dot{Q}_L$) loads, which depend on internal heat sources ($\dot{Q}_{I,S}$ and $\dot{Q}_{I,L}$), and environmental heat transfers. Assume the zone has highly turbulent mixed ventilation such that the internal space is at a uniform dry bulb temperature and humidity state (indicated by state 5), which is at a higher energy condition (due to the sensible and latent cooling load heat transfers) than the air that enters the zone. Exhaust air at state 5 is drawn by the exhaust fan and passed down an air passage until it reaches state 6. The air is passed to the environment and a portion could be mixed with the inlet air as described earlier. No air bypasses the cooling coil and there are no known air leakages in the circuit. Assume all air ducts and fans to be adiabatic. Ignore fan work for now.

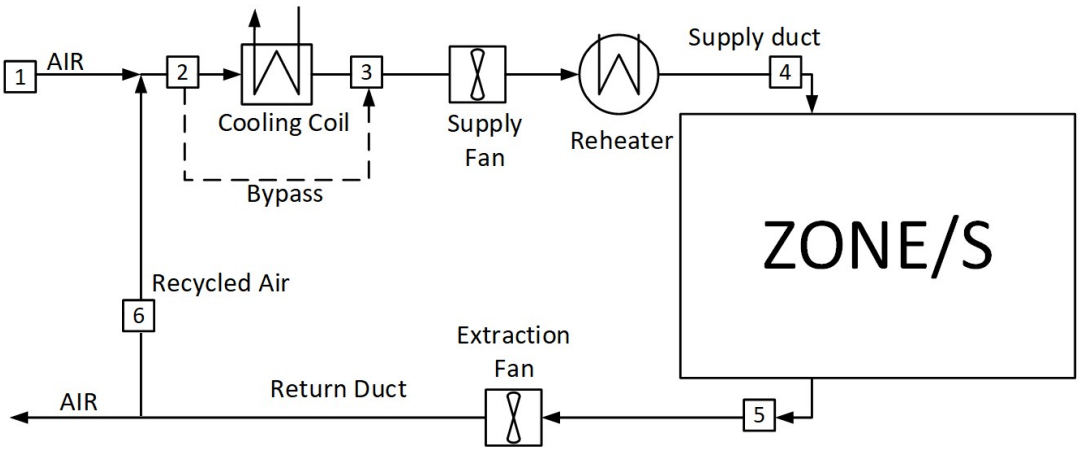


Figure 1 Air conditioning system

The entire installation is at a high altitude with an atmospheric pressure (P_t) of 90 kPa. The outdoor air state is 35°C dry-bulb (t_1) and 27°C wet bulb (b_1). Outdoor is drawn in a constant volume flow rate ($\dot{V}_1$) of 1.5 m^3/s. For simplicity, use a constant cooling rate ($\dot{Q}_{CC}$) of 50 kW and relative humidity directly after the cooling coil (ϕ_3) of 0.8. (In reality these values will depend on the operating state and the characteristics of the cooling coil and will not be constant, as will be dealt with in a future portion of MLR780.) The reheater is switched off. For purposes of this assignment the heat ingression to the zone can be described as follows:

$\dot{Q}_S = \dot{Q}_{I,S} + \frac{t_1 - t_5}{R_t}$ and $\dot{Q}_L = \dot{Q}_{I,L}$

With R_t = 0.001 K/W being the thermal resistance of the building structure and $\dot{Q}_{I,S}$ = 2 kW and $\dot{Q}_{I,L}$= 700 W. (These values will be calculated in more detail in a future part of this module).

Where applicable, evaluate (as an estimation) the specific heat quantities at the inlet state of relevant device. **Make sure you use CoolProp 6.4.1 or higher.**

Analysis:

First consider a case with no air recirculation ($\dot{m}_6$=0 kg/s):

1) Calculate the missing properties of: dry bulb temperature (t), specific enthalpy (h) of the moist air and humidity ratio (W) at state 1 to 6. Show all relevant equations, calculation methods and any other relevant thermodynamic property or calculated quantity. Iterative solution is required for state 5. Do not list result of your iterations, only the converged answer set. Give a summary of your answers as follows:

State:	**1**	**2**	**3**	**4**	**5**	**6**
t [°C]						
h [J/kg]						
W [kg/kg]						

Specifically comment about the conditions at state 5.

2) Specifically document the calculation (equations and method) of the following quantities: Also summarise final results in table form as follows:

Device:	**Cooling Coil**	**Supply Fan**	**Reheater**	**Zone**	**Return Duct**
Sensible heat rate [W]					
Latent heat rate [W]					
Total heat rate [W]					
Moisture addition (+) or removal (-) [kg/s]					

Now consider an increased air flow rate in the zone, while maintaining the same outdoor air mass rate as before, but with $\dot{m}_6$= 2 kg/s.

3) Repeat question 1. Still summarise your calculated properties in table form as in question 1, but only document your calculation methods for state 2. Because of presence of recycled air: in addition to the iterative calculations needed for state 5 (as in question 1), an additional higher-level Iteration scheme is needed to solve the whole set of thermodynamic states. Do not list results of your iterations, but only the final converged conditions of the entire system.

Now analyse several cases for $\dot{m}_6$ (between 0 and 50 kg/s):

4) Tabulate and plot the effect of $\dot{m}_6$ on t_5 and ϕ_5. Include data point markers in the plots. Full documentation of calculations is not required.
5) Discuss the possible impact of altering $\dot{m}_6$ on the internal condition in the zone.

REMEMBER TO INDICATE ON YOUR FIRST SUBMISSION PAGE HOW MUCH TIME YOU SPENT ON THE ASSIGNMENT